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## Introduction

**Continual learning** asks whether a neural network can learn a sequence of tasks without forgetting what it learned before. In practice, networks fail catastrophically: gradient descent on a new task overwrites weights that encoded prior knowledge, collapsing old-task accuracy within just a few update steps.

The field has produced three families of fixes — **regularization** (penalize changes to important weights), **replay** (store and re-train on old examples), and **architecture** methods (partition capacity per task) — but none solve the problem in a principled, parameter-efficient way. Crucially, almost all operate in **Euclidean parameter space**, which is not reparameterization-invariant and gives an unreliable notion of which weights matter.

## FOPNG: Fisher-Orthogonal Projected Natural Gradient Descent

### Fisher Information and Natural Gradient

The **Fisher information matrix**  $\mathcal{F}_\theta$  measures how sensitively the model's output distribution responds to each parameter:

$$\mathcal{F}_\theta = \mathbb{E}[\nabla_\theta \log p(y|x; \theta) \nabla_\theta \log p(y|x; \theta)^\top] \quad (1)$$

Standard gradient descent ignores this curvature. **Natural gradient descent** corrects it by preconditioning with  $\mathcal{F}_\theta^{-1}$ , yielding updates that are equal-sized in distribution space and reparameterization-invariant:

$$\Delta\theta_{\text{nat}} = -\eta \mathcal{F}_\theta^{-1} \nabla_\theta L \quad (2)$$

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## Orthogonal Projection

After each task  $k$ , we store its gradient  $g_k$  in a memory matrix  $G = [g_1 \mid \dots \mid g_{t-1}]$ . New updates are projected onto the **Fisher-orthogonal complement** of  $G$  — directions that cannot alter prior task outputs:

$$\Delta\theta = -\eta (I - G(G^\top \mathcal{F}_\theta G)^{-1} G^\top \mathcal{F}_\theta) \mathcal{F}_\theta^{-1} \nabla_\theta L_t \quad (3)$$

The term  $G(G^\top \mathcal{F}_\theta G)^{-1} G^\top \mathcal{F}_\theta$  is a projection that reads off and removes any component of the new gradient that would disturb old tasks. What remains is a geometrically hard-constrained update.

## Hypernetworks

A **hypernetwork**  $h(\cdot; \phi)$  generates the weights of a target network rather than making predictions itself. Each task  $t$  receives a learned embedding  $z_t$ , producing task-specific weights on the fly:

$$\theta_t = h(z_t; \phi) \quad (4)$$

The shared parameters  $\phi$  do not grow with the number of tasks, and each task gets a fully specialized network by varying  $z_t$ . Applying FOPNG to  $\phi$  means the Fisher matrix and gradient memory are maintained only for this compact set of parameters — not the larger target network.

For very large target networks, **chunking** keeps  $\phi$  tractable: the hypernetwork generates target weights in small blocks, each conditioned on  $z_t$  and a chunk index, rather than all at once.

## Benchmarks

**Split-MNIST** splits MNIST into 5 binary tasks (e.g. 0 vs 1, then 2 vs 3). The standard entry-level benchmark for sequential learning.

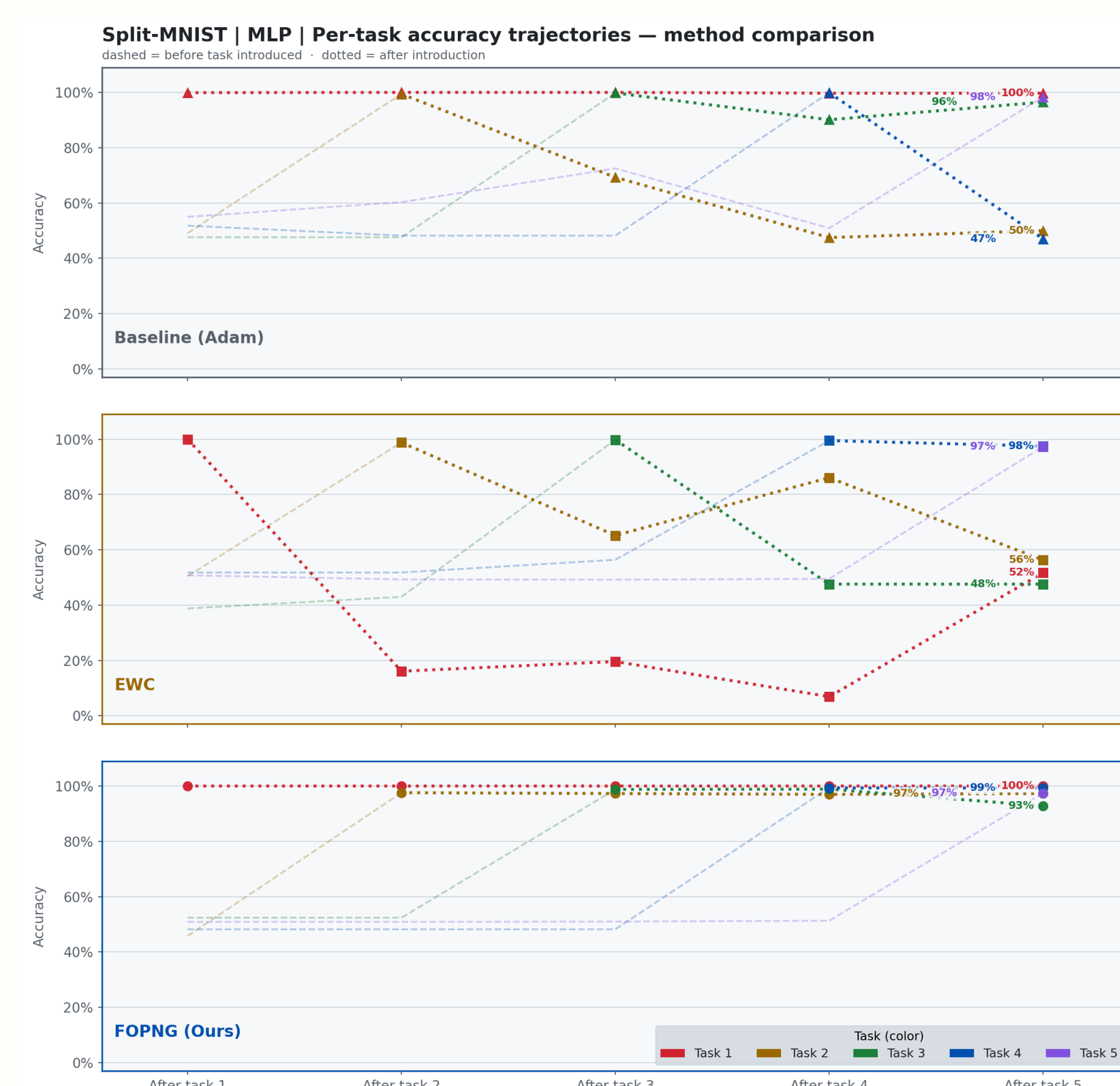


Figure 1: Standard MLP performance comparison. Notice the severe performance degradation in Baseline (Adam) and partial forgetting in EWC.

**Permuted-MNIST** applies a different random pixel permutation per task, all sharing the same label space. Tests long-horizon retention over 10+ tasks.

**Split-CIFAR-10** and **Split-CIFAR-100** apply the split protocol to natural images: 5 tasks over 10 classes, and 20 tasks over 100 classes respectively. Higher visual complexity and more tasks make these substantially harder.

## Evaluation Metrics

We report **average accuracy** across all tasks after training, **backward transfer** (how much new training degrades old tasks), **forward transfer** (whether prior tasks help new ones), and **per-task accuracy trajectories** to visualize forgetting dynamics over time.

## Summary and conclusions

In this thesis project, we successfully formulated and evaluated a novel continual learning framework combining Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) with Hypernetworks.

Our key findings include:

- **Geometric Orthogonality:** Enforcing orthogonality in the Fisher-Riemannian manifold is vastly superior to Euclidean space.
- **Synergy with Hypernetworks:** Hypernetworks serve as an excellent substrate for FOPNG, localizing complex computations to parameters  $\phi$ .
- **State-of-the-Art Performance:** Our approach achieves marked improvements on Split-MNIST compared to baselines.

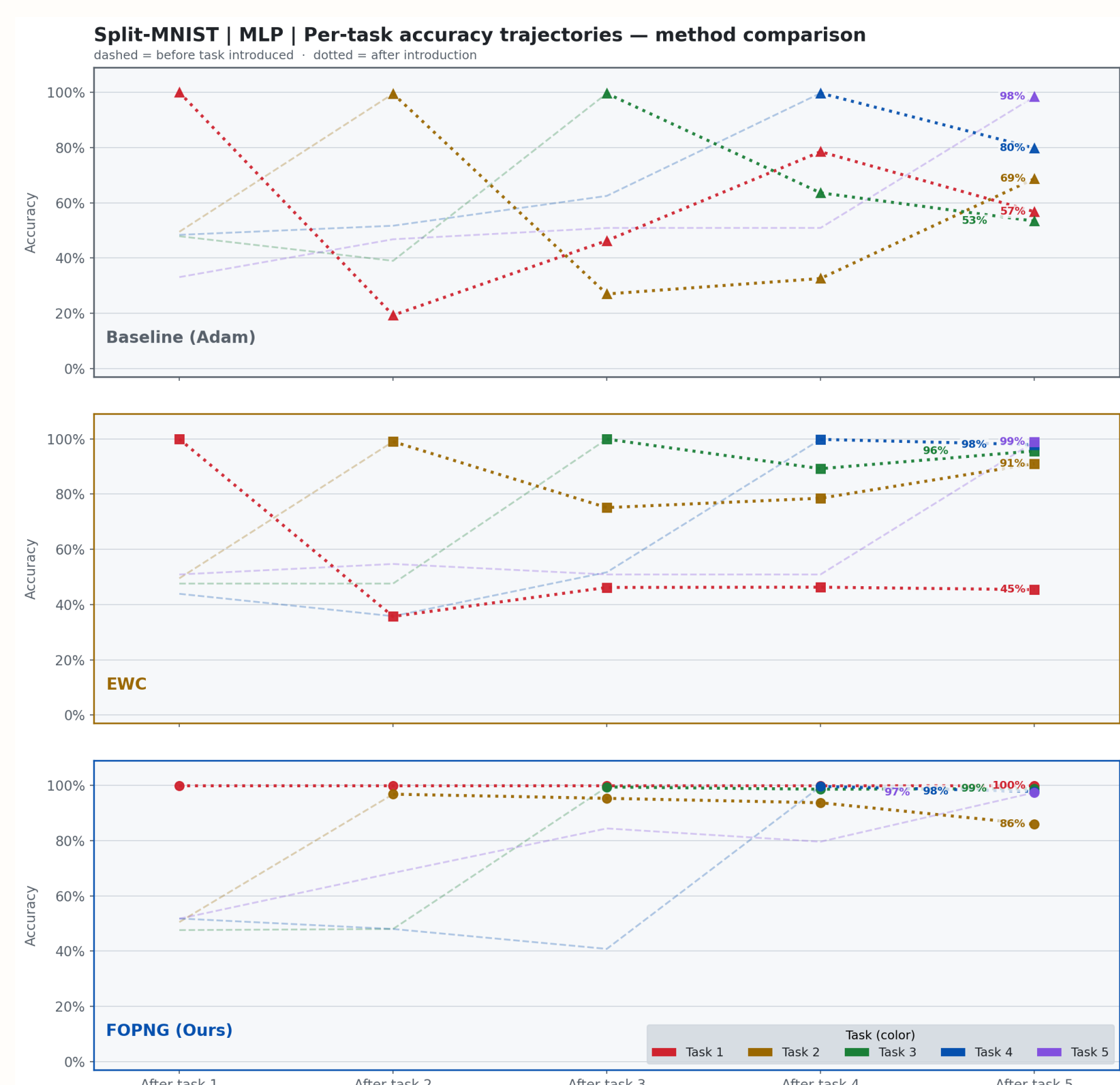


Figure 2: Hypernetwork performance on Split-MNIST. FOPNG successfully maintains high accuracy across all sequential tasks without catastrophic forgetting.